

Section 1: Preprocessing & Text Cleaning

Regex Cleaning

- Pattern: `re.sub(r' [^\w\s]', '', text)`
- Removes:
 - Punctuation (! ? , .)
 - Emojis
- **! Issue:** Removes sentiment indicators
 - Example: "Great service 😊" → becomes neutral
- **Impact:** Loss of emotional context → poor sentiment analysis

Extracting US Phone Numbers

Regex Pattern:

`\( \d{3} \) \s \d{3} - \d{4}`

Explanation:

- `\( \d{3} \)` → Area code (e.g., (555))
- `\s` → Space
- `\d{3} - \d{4}` → Number format

Ignores international numbers because:

- No + allowed
- Strict US format

Lowercasing & NER

- Lowercasing removes capitalization clues
- Example:
 - "Apple launched iPhone" → entity (Organization)
 - "apple launched iPhone" → may not detect entity

Impact: Reduced NER accuracy

Stopword Removal Issues

- Words like: **not, no, very**
- Example:
 - "I do not like this product"
 - After removal → "I do like this product" ✗

Solution:

- Keep negation words
- Use negation tagging (e.g., `not_like`)

Section 2: Normalization (Stemming vs Lemmatization)

Stemming vs Lemmatization

Feature	Stemming	Lemmatization
Method	Rule-based	Dictionary-based
Output	May be incorrect	Correct root word
Speed	Fast	Slower

Example: "better"

- Stemmer → **better** (no proper root)
- Lemmatizer → **good**

Reason:

- Stemming cuts words blindly
- Lemmatization uses meaning

Lemmatization Failure ("leaves")

- Without POS tag → remains "leaves"
- Because meaning unclear

- Provide POS:
 - Noun → leaf
 - Verb → leave

Stemming for Search Engines

- Preferred for large datasets (e.g., Reddit)

Reasons:

- Faster
- Handles variations
- Improves recall

Stemming Collision Problem

Words:

- university
- universe
- universal

→ All become **univers**

Issue:

- Different meanings merged
- Poor topic classification

Lemmatization Limitation

- Depends on dictionary
- Cannot handle:
 - Slang (lit, bro)
 - New words
 - Misspellings

Section 3: POS Tagging & NER

Context-based POS Tagging

Sentence:

"I will book the flight after I read this book"

- "book" (after will) → Verb
- "book" (after this) → Noun

Uses surrounding words for context

NER Error: "Apple"

Sentence: "I ate a green Apple"

Error: Tagged as Organization

Cause:

- Lowercasing
- Lack of context

Solution:

- Preserve casing
- Use context-aware models

Extracting Actions

Use POS tags:

- Verbs (VB, VBD, VBG)

Why:

- Actions = verbs
- Example: ordered, clicked, cancelled

Historical Text Problem

- Old texts have different capitalization

Result:

- Poor NER accuracy

Solution:

- Fine-tune model
- Normalize text

Multi-word Entities

Example: "New York City"

Represent as:

- **Single token**

Advantage:

- Preserves meaning
- Avoids fragmentation

Section 4: Bag of Words (BoW)

Vectorization Example

Vocabulary:

['apple', 'banana', 'eat', 'i', 'like']

Sentence:

"I eat the apple, I like the apple"

Vector:

[2, 0, 1, 2, 1]

"the" ignored (not in vocab)

Reducing Dimensionality

Techniques:

- Stopword removal
- Stemming / Lemmatization
- Remove rare words

BoW Limitation (Order Ignored)

Sentences:

- "Dog bit man"
- "Man bit dog"

Same vector → **✗** meaning lost

Frequency $\neq$ Importance

- High frequency $\neq$ important
- Example: "server" appears many times

Use:

- **TF-IDF** for better weighting

Unknown Words

- Words not in vocabulary → ignored

Impact:

- Model uses limited information
- Accuracy may reduce

◆ Synthesis: NLP Pipeline

4-Step Pipeline

1. Text Cleaning

- Remove noise, punctuation

2. POS Tagging

- Identify nouns, verbs, etc.

Needed before filtering

3. Stopword Removal

- Remove common words
- Keep negations

4. Vectorization (BoW)

- Convert text → numbers

Why Order Matters

- POS before stopwords → keeps important words
- Stopwords before BoW → reduces size
- Clean text → improves accuracy